# class08_mini_project

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### 1. Background

In today's class, we will apply the methods and techniques clustering and PCA to help make sense of a real world breast cancer FNA biopsy data set.

radius: mean of distances from nucleus center to points on the perimeter; texture: a measure of nucleus roughness taken from the standard deviation of gray-scale values; perimeter: total boundary length of the nucleus, area: total area of the nucleus; smoothness: local variation in radius lengths, i.e. how "bumpy" the edge is; compactness: measures how circular vs. irregular the shape is; concavity: how deeply indented, and symmetry: how symmetric the nucleus is

## 2. Exploratory Data Analysis

```r
library(ggplot2)
# Save your input data file into your Project directory
fna.data <- "WisconsinCancer.csv"

# Complete the following code to input the data and store as wisc.df
wisc.df <- read.csv(fna.data, row.names=1)

head(wisc.df, 4)
```

```
         diagnosis radius_mean texture_mean perimeter_mean area_mean
842302           M       17.99        10.38         122.80    1001.0
842517           M       20.57        17.77         132.90    1326.0
84300903         M       19.69        21.25         130.00    1203.0
84348301         M       11.42        20.38          77.58     386.1
         smoothness_mean compactness_mean concavity_mean concave.points_mean
842302           0.11840          0.27760         0.3001             0.14710
842517           0.08474          0.07864         0.0869             0.07017
84300903         0.10960          0.15990         0.1974             0.12790
84348301         0.14250          0.28390         0.2414             0.10520
         symmetry_mean fractal_dimension_mean radius_se texture_se perimeter_se
842302          0.2419                0.07871    1.0950     0.9053        8.589
842517          0.1812                0.05667    0.5435     0.7339        3.398
84300903        0.2069                0.05999    0.7456     0.7869        4.585
84348301        0.2597                0.09744    0.4956     1.1560        3.445
         area_se smoothness_se compactness_se concavity_se concave.points_se
842302    153.40      0.006399        0.04904      0.05373           0.01587
842517     74.08      0.005225        0.01308      0.01860           0.01340
84300903   94.03      0.006150        0.04006      0.03832           0.02058
84348301   27.23      0.009110        0.07458      0.05661           0.01867
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
```

```
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
```

Make sure to remove the first `diagnosis` column – I don't want to use this for my machine learning models. We'll use this later to compare our results to the expected diagnosis.

```r
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
head(wisc.data)
```

```
         radius_mean texture_mean perimeter_mean area_mean smoothness_mean
842302         17.99        10.38         122.80    1001.0         0.11840
842517         20.57        17.77         132.90    1326.0         0.08474
84300903       19.69        21.25         130.00    1203.0         0.10960
84348301       11.42        20.38          77.58     386.1         0.14250
84358402       20.29        14.34         135.10    1297.0         0.10030
843786         12.45        15.70          82.57     477.1         0.12780
         compactness_mean concavity_mean concave.points_mean symmetry_mean
842302            0.27760         0.3001             0.14710        0.2419
842517            0.07864         0.0869             0.07017        0.1812
84300903          0.15990         0.1974             0.12790        0.2069
84348301          0.28390         0.2414             0.10520        0.2597
84358402          0.13280         0.1980             0.10430        0.1809
843786            0.17000         0.1578             0.08089        0.2087
         fractal_dimension_mean radius_se texture_se perimeter_se area_se
842302                  0.07871    1.0950     0.9053        8.589  153.40
842517                  0.05667    0.5435     0.7339        3.398   74.08
84300903                0.05999    0.7456     0.7869        4.585   94.03
84348301                0.09744    0.4956     1.1560        3.445   27.23
84358402                0.05883    0.7572     0.7813        5.438   94.44
843786                  0.07613    0.3345     0.8902        2.217   27.19
         smoothness_se compactness_se concavity_se concave.points_se
842302        0.006399        0.04904      0.05373           0.01587
842517        0.005225        0.01308      0.01860           0.01340
```

```
84300903      0.006150        0.04006      0.03832           0.02058
84348301      0.009110        0.07458      0.05661           0.01867
84358402      0.011490        0.02461      0.05688           0.01885
843786        0.007510        0.03345      0.03672           0.01137
         symmetry_se fractal_dimension_se radius_worst texture_worst
842302       0.03003             0.006193        25.38         17.33
842517       0.01389             0.003532        24.99         23.41
84300903     0.02250             0.004571        23.57         25.53
84348301     0.05963             0.009208        14.91         26.50
84358402     0.01756             0.005115        22.54         16.67
843786       0.02165             0.005082        15.47         23.75
         perimeter_worst area_worst smoothness_worst compactness_worst
842302            184.60     2019.0           0.1622            0.6656
842517            158.80     1956.0           0.1238            0.1866
84300903          152.50     1709.0           0.1444            0.4245
84348301           98.87      567.7           0.2098            0.8663
84358402          152.20     1575.0           0.1374            0.2050
843786            103.40      741.6           0.1791            0.5249
         concavity_worst concave.points_worst symmetry_worst
842302            0.7119               0.2654         0.4601
842517            0.2416               0.1860         0.2750
84300903          0.4504               0.2430         0.3613
84348301          0.6869               0.2575         0.6638
84358402          0.4000               0.1625         0.2364
843786            0.5355               0.1741         0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

Q1. How many observations are in this dataset?

```r
# Create diagnosis vector for later
diagnosis <- wisc.df$diagnosis
head(diagnosis)
```

```
[1] "M" "M" "M" "M" "M" "M"
```

```r
dim(wisc.data)
```

```
[1] 569  30
```

> ans: 569

Q2. How many of the observations have a malignant diagnosis?

```r
#How many malignant diagnosis are in the dataset?
wisc.df$diagnosis == "M"
```

```
  [1]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
 [13]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE
 [25]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
 [37]  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE
 [49] FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE FALSE FALSE
 [61] FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE FALSE
 [73]  TRUE  TRUE FALSE  TRUE FALSE  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE
 [85] FALSE  TRUE  TRUE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE  TRUE
 [97] FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
[109]  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE  TRUE
[121] FALSE  TRUE  TRUE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE FALSE  TRUE
[133]  TRUE FALSE  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE FALSE
[145] FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[157]  TRUE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE FALSE FALSE  TRUE
[169]  TRUE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
[181]  TRUE  TRUE  TRUE FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE
[193] FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE
[205] FALSE  TRUE FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE
[217] FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
[229] FALSE  TRUE  TRUE FALSE FALSE  TRUE FALSE FALSE  TRUE  TRUE FALSE  TRUE
[241] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
[253]  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
[265]  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE
[277] FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE
[289] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
[301]  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[313] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE  TRUE
[325] FALSE FALSE FALSE FALSE  TRUE  TRUE  TRUE FALSE FALSE FALSE FALSE  TRUE
[337] FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
```

```
[349] FALSE FALSE FALSE  TRUE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE
[361] FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE  TRUE  TRUE FALSE
[373]  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE
[385] FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE  TRUE FALSE FALSE
[397] FALSE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[409]  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE FALSE
[421] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE
[433]  TRUE  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE
[445]  TRUE FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE FALSE FALSE
[457] FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE
[469]  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE
[481] FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE FALSE  TRUE FALSE FALSE
[493]  TRUE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE FALSE  TRUE FALSE  TRUE
[505] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE FALSE  TRUE FALSE  TRUE FALSE
[517]  TRUE  TRUE FALSE FALSE FALSE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE
[529] FALSE FALSE FALSE FALSE FALSE  TRUE FALSE  TRUE  TRUE FALSE FALSE FALSE
[541] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[553] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE  TRUE  TRUE
[565]  TRUE  TRUE  TRUE  TRUE FALSE
```

```r
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

```r
#Alternative way...
head(table(wisc.df$diagnosis))
```

```
  B   M
357 212
```

```
> ans: 212
```

Q3. How many variables/features in the data are suffixed with _mean?

```r
head(wisc.data)
```

```
         radius_mean texture_mean perimeter_mean area_mean smoothness_mean
842302         17.99        10.38         122.80    1001.0         0.11840
```

| | | | | | |
|---|---|---|---|---|---|
| 842517 | 20.57 | 17.77 | 132.90 | 1326.0 | 0.08474 |
| 84300903 | 19.69 | 21.25 | 130.00 | 1203.0 | 0.10960 |
| 84348301 | 11.42 | 20.38 | 77.58 | 386.1 | 0.14250 |
| 84358402 | 20.29 | 14.34 | 135.10 | 1297.0 | 0.10030 |
| 843786 | 12.45 | 15.70 | 82.57 | 477.1 | 0.12780 |

| | compactness_mean | concavity_mean | concave.points_mean | symmetry_mean |
|---|---|---|---|---|
| 842302 | 0.27760 | 0.3001 | 0.14710 | 0.2419 |
| 842517 | 0.07864 | 0.0869 | 0.07017 | 0.1812 |
| 84300903 | 0.15990 | 0.1974 | 0.12790 | 0.2069 |
| 84348301 | 0.28390 | 0.2414 | 0.10520 | 0.2597 |
| 84358402 | 0.13280 | 0.1980 | 0.10430 | 0.1809 |
| 843786 | 0.17000 | 0.1578 | 0.08089 | 0.2087 |

| | fractal_dimension_mean | radius_se | texture_se | perimeter_se | area_se |
|---|---|---|---|---|---|
| 842302 | 0.07871 | 1.0950 | 0.9053 | 8.589 | 153.40 |
| 842517 | 0.05667 | 0.5435 | 0.7339 | 3.398 | 74.08 |
| 84300903 | 0.05999 | 0.7456 | 0.7869 | 4.585 | 94.03 |
| 84348301 | 0.09744 | 0.4956 | 1.1560 | 3.445 | 27.23 |
| 84358402 | 0.05883 | 0.7572 | 0.7813 | 5.438 | 94.44 |
| 843786 | 0.07613 | 0.3345 | 0.8902 | 2.217 | 27.19 |

| | smoothness_se | compactness_se | concavity_se | concave.points_se |
|---|---|---|---|---|
| 842302 | 0.006399 | 0.04904 | 0.05373 | 0.01587 |
| 842517 | 0.005225 | 0.01308 | 0.01860 | 0.01340 |
| 84300903 | 0.006150 | 0.04006 | 0.03832 | 0.02058 |
| 84348301 | 0.009110 | 0.07458 | 0.05661 | 0.01867 |
| 84358402 | 0.011490 | 0.02461 | 0.05688 | 0.01885 |
| 843786 | 0.007510 | 0.03345 | 0.03672 | 0.01137 |

| | symmetry_se | fractal_dimension_se | radius_worst | texture_worst |
|---|---|---|---|---|
| 842302 | 0.03003 | 0.006193 | 25.38 | 17.33 |
| 842517 | 0.01389 | 0.003532 | 24.99 | 23.41 |
| 84300903 | 0.02250 | 0.004571 | 23.57 | 25.53 |
| 84348301 | 0.05963 | 0.009208 | 14.91 | 26.50 |
| 84358402 | 0.01756 | 0.005115 | 22.54 | 16.67 |
| 843786 | 0.02165 | 0.005082 | 15.47 | 23.75 |

| | perimeter_worst | area_worst | smoothness_worst | compactness_worst |
|---|---|---|---|---|
| 842302 | 184.60 | 2019.0 | 0.1622 | 0.6656 |
| 842517 | 158.80 | 1956.0 | 0.1238 | 0.1866 |
| 84300903 | 152.50 | 1709.0 | 0.1444 | 0.4245 |
| 84348301 | 98.87 | 567.7 | 0.2098 | 0.8663 |
| 84358402 | 152.20 | 1575.0 | 0.1374 | 0.2050 |
| 843786 | 103.40 | 741.6 | 0.1791 | 0.5249 |

| | concavity_worst | concave.points_worst | symmetry_worst |
|---|---|---|---|
| 842302 | 0.7119 | 0.2654 | 0.4601 |
| 842517 | 0.2416 | 0.1860 | 0.2750 |

```
84300903          0.4504                0.2430          0.3613
84348301          0.6869                0.2575          0.6638
84358402          0.4000                0.1625          0.2364
843786            0.5355                0.1741          0.3985
         fractal_dimension_worst
842302                   0.11890
842517                   0.08902
84300903                 0.08758
84348301                 0.17300
84358402                 0.07678
843786                   0.12440
```

```r
meannames<-grep("_mean", colnames(wisc.df), value = TRUE)
meannames2<-grep("_mean", colnames(wisc.df), value = TRUE)

#`grep()` searches for a pattern in x
length(meannames2)
```

```
[1] 10
```

```
  > ans: 10
```

### 3. PCA

Practical PCA issue is scaling… scale and center our data before PCA, otherwise the data is going to be unfairly influenced because your data is going to be huge. (eg: if you measured one point in mm versus another in miles)

*prcomp(function for pca)(x = required argument, scale and center = optional argument)*

The main function here is `prcomp()` and we want to make sure we set the optional arguments to `scale=true`

```r
wisc.pr <- prcomp(wisc.data, scale = TRUE)
summary(wisc.pr)
```

```
Importance of components:
                          PC1    PC2     PC3     PC4     PC5     PC6     PC7
Standard deviation     3.6444 2.3857 1.67867 1.40735 1.28403 1.09880 0.82172
Proportion of Variance 0.4427 0.1897 0.09393 0.06602 0.05496 0.04025 0.02251
Cumulative Proportion  0.4427 0.6324 0.72636 0.79239 0.84734 0.88759 0.91010
```

```
                           PC8    PC9    PC10   PC11    PC12    PC13    PC14
Standard deviation     0.69037 0.6457 0.59219 0.5421 0.51104 0.49128 0.39624
Proportion of Variance 0.01589 0.0139 0.01169 0.0098 0.00871 0.00805 0.00523
Cumulative Proportion  0.92598 0.9399 0.95157 0.9614 0.97007 0.97812 0.98335
                          PC15    PC16    PC17    PC18    PC19    PC20   PC21
Standard deviation     0.30681 0.28260 0.24372 0.22939 0.22244 0.17652 0.1731
Proportion of Variance 0.00314 0.00266 0.00198 0.00175 0.00165 0.00104 0.0010
Cumulative Proportion  0.98649 0.98915 0.99113 0.99288 0.99453 0.99557 0.9966
                          PC22    PC23   PC24    PC25    PC26    PC27    PC28
Standard deviation     0.16565 0.15602 0.1344 0.12442 0.09043 0.08307 0.03987
Proportion of Variance 0.00091 0.00081 0.0006 0.00052 0.00027 0.00023 0.00005
Cumulative Proportion  0.99749 0.99830 0.9989 0.99942 0.99969 0.99992 0.99997
                          PC29    PC30
Standard deviation     0.02736 0.01153
Proportion of Variance 0.00002 0.00000
Cumulative Proportion  1.00000 1.00000
```

```r
#Cumulative proportion calculates the POV for previous PCAs

# Check column means and standard deviations
colMeans(wisc.data)
```

```
            radius_mean           texture_mean         perimeter_mean
           1.412729e+01           1.928965e+01           9.196903e+01
              area_mean        smoothness_mean       compactness_mean
           6.548891e+02           9.636028e-02           1.043410e-01
         concavity_mean    concave.points_mean          symmetry_mean
           8.879932e-02           4.891915e-02           1.811619e-01
 fractal_dimension_mean              radius_se             texture_se
           6.279761e-02           4.051721e-01           1.216853e+00
           perimeter_se                area_se          smoothness_se
           2.866059e+00           4.033708e+01           7.040979e-03
         compactness_se           concavity_se      concave.points_se
           2.547814e-02           3.189372e-02           1.179614e-02
            symmetry_se   fractal_dimension_se           radius_worst
           2.054230e-02           3.794904e-03           1.626919e+01
          texture_worst        perimeter_worst             area_worst
           2.567722e+01           1.072612e+02           8.805831e+02
       smoothness_worst      compactness_worst        concavity_worst
           1.323686e-01           2.542650e-01           2.721885e-01
   concave.points_worst         symmetry_worst fractal_dimension_worst
           1.146062e-01           2.900756e-01           8.394582e-02
```

```r
apply(wisc.data,2,sd)
```

```
           radius_mean           texture_mean         perimeter_mean
          3.524049e+00           4.301036e+00           2.429898e+01
             area_mean        smoothness_mean       compactness_mean
          3.519141e+02           1.406413e-02           5.281276e-02
        concavity_mean    concave.points_mean          symmetry_mean
          7.971981e-02           3.880284e-02           2.741428e-02
 fractal_dimension_mean              radius_se             texture_se
          7.060363e-03           2.773127e-01           5.516484e-01
          perimeter_se                area_se          smoothness_se
          2.021855e+00           4.549101e+01           3.002518e-03
        compactness_se           concavity_se      concave.points_se
          1.790818e-02           3.018606e-02           6.170285e-03
           symmetry_se   fractal_dimension_se            radius_worst
          8.266372e-03           2.646071e-03           4.833242e+00
         texture_worst        perimeter_worst              area_worst
          6.146258e+00           3.360254e+01           5.693570e+02
      smoothness_worst      compactness_worst        concavity_worst
          2.283243e-02           1.573365e-01           2.086243e-01
  concave.points_worst         symmetry_worst fractal_dimension_worst
          6.573234e-02           6.186747e-02           1.806127e-02
```

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

```
>ans: 44.27%
```

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

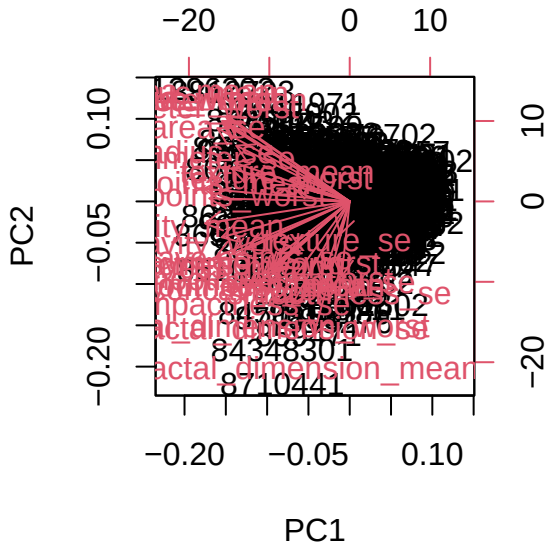
```
>ans: 3
```

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

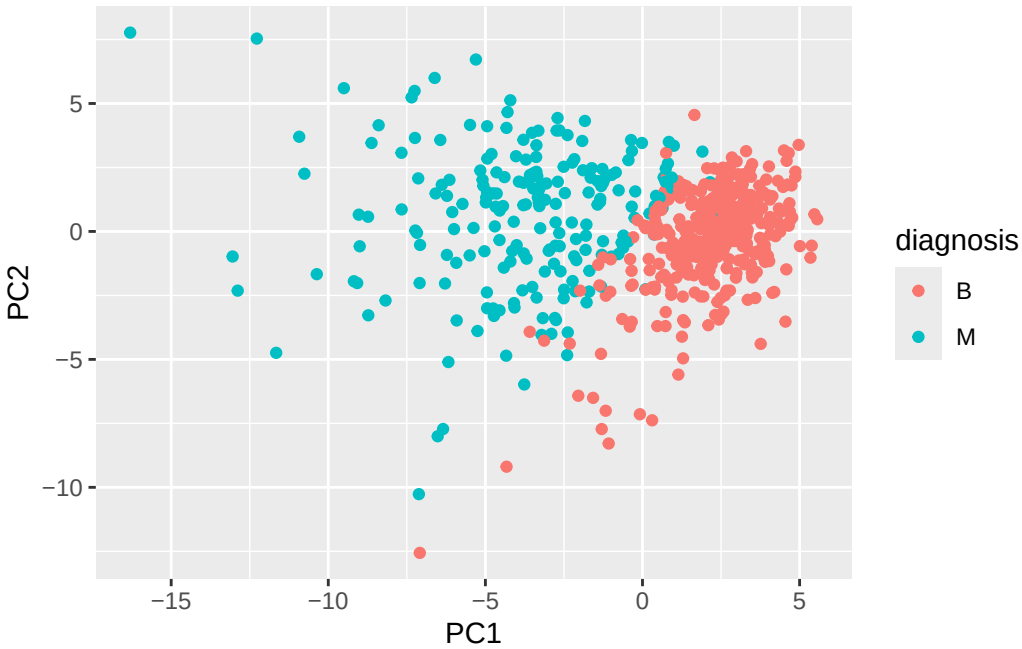
```
>ans: 7
```

```r
library(ggplot2)
```

```r
biplot(wisc.pr)
```




```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```




Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

```
> ans: This plot is very clustered together and not easy to read.
```

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
> ans: This plot is much easier to read because it narrows down groups into visually manag
```

*More on PCA score plots…*

This graph shows that cells with similar characteristics will cluster.

```
library("stats")
```

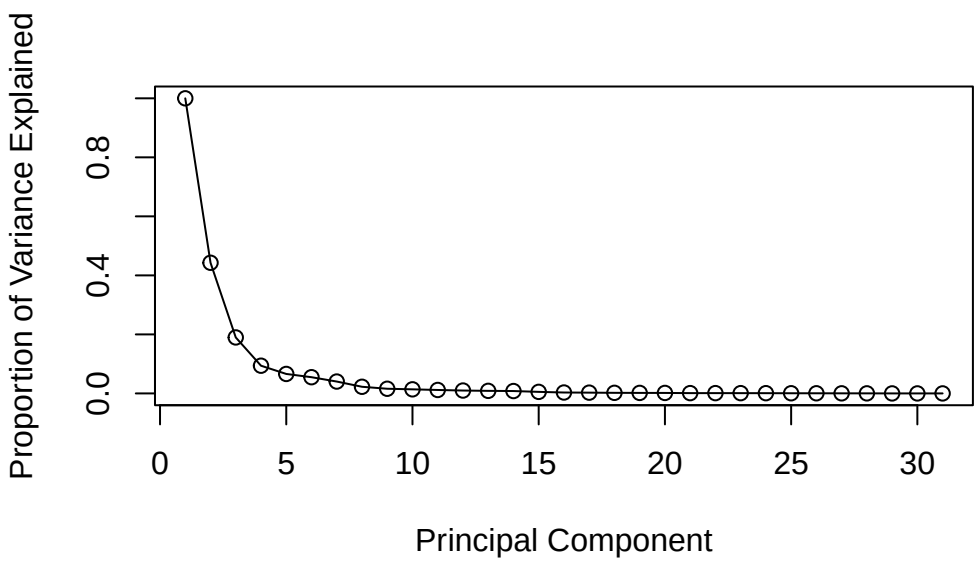
```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

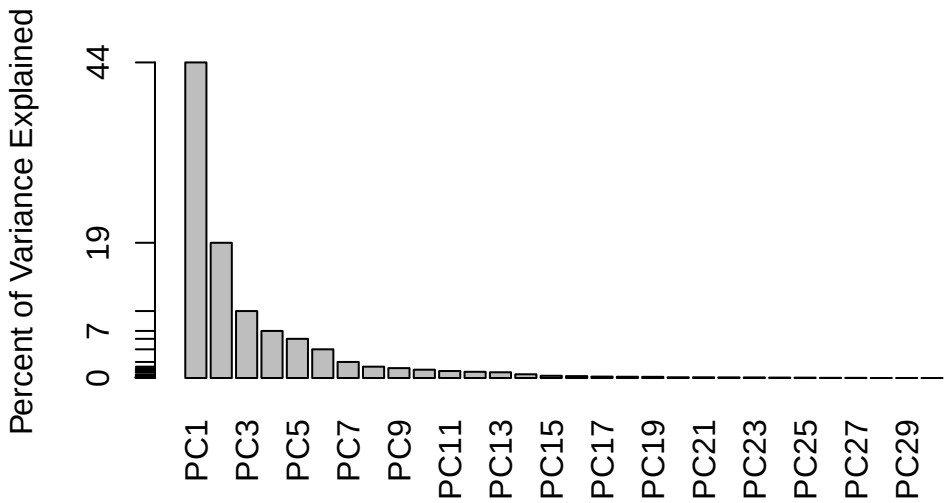
```r
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)
pve
```

```
 [1] 4.427203e-01 1.897118e-01 9.393163e-02 6.602135e-02 5.495768e-02
 [6] 4.024522e-02 2.250734e-02 1.588724e-02 1.389649e-02 1.168978e-02
[11] 9.797190e-03 8.705379e-03 8.045250e-03 5.233657e-03 3.137832e-03
[16] 2.662093e-03 1.979968e-03 1.753959e-03 1.649253e-03 1.038647e-03
[21] 9.990965e-04 9.146468e-04 8.113613e-04 6.018336e-04 5.160424e-04
[26] 2.725880e-04 2.300155e-04 5.297793e-05 2.496010e-05 4.434827e-06
```

```r
# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```




```r
barplot(pve, ylab = "Percent of Variance Explained",
     names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
      axis(2, at=pve, labels=round(pve,2)*100 )
```

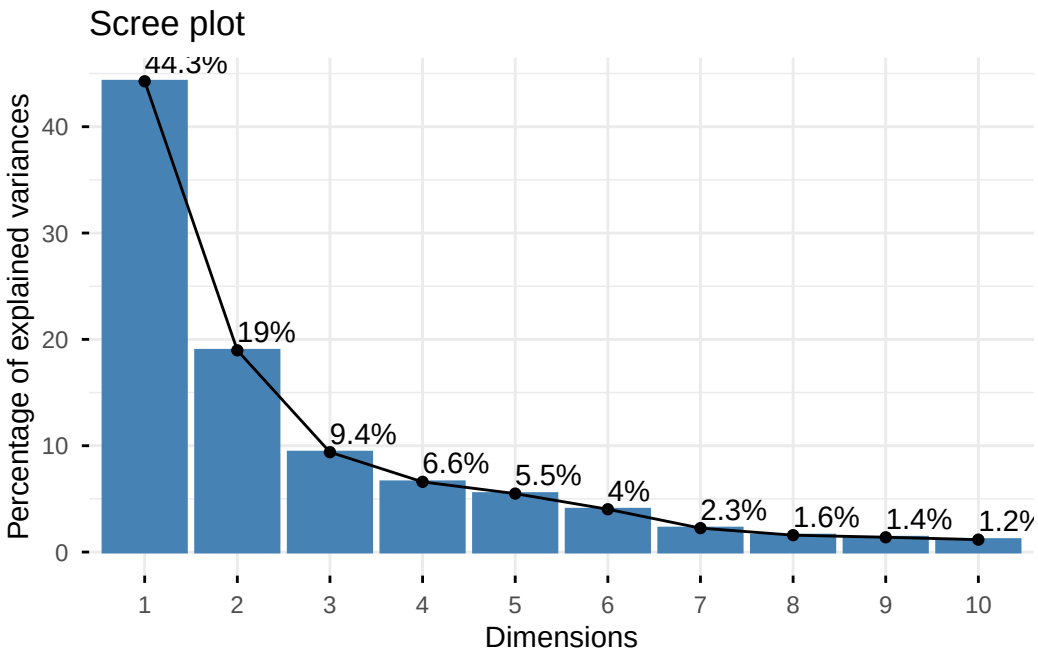



```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

```
Welcome to factoextra!

Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/
guide-to-principal-component-methods-in-r/
```

```
fviz_eig(wisc.pr, addlabels = TRUE)
```




```
wisc.pr$rotation[,1]
```

```
            radius_mean           texture_mean         perimeter_mean
            -0.21890244            -0.10372458            -0.22753729
              area_mean        smoothness_mean       compactness_mean
            -0.22099499            -0.14258969            -0.23928535
         concavity_mean     concave.points_mean          symmetry_mean
            -0.25840048            -0.26085376            -0.13816696
 fractal_dimension_mean              radius_se             texture_se
            -0.06436335            -0.20597878            -0.01742803
           perimeter_se                area_se          smoothness_se
            -0.21132592            -0.20286964            -0.01453145
         compactness_se           concavity_se      concave.points_se
            -0.17039345            -0.15358979            -0.18341740
            symmetry_se    fractal_dimension_se           radius_worst
            -0.04249842            -0.10256832            -0.22799663
          texture_worst        perimeter_worst             area_worst
            -0.10446933            -0.23663968            -0.22487053
       smoothness_worst       compactness_worst        concavity_worst
            -0.12795256            -0.21009588            -0.22876753
   concave.points_worst          symmetry_worst fractal_dimension_worst
            -0.25088597            -0.12290456            -0.13178394
```

Q9. For the first principal component, what is the component of the loading vector (i.e. wisc.pr$rotation[,1]) for the feature concave.points_mean? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
> ans:  -0.26085376. No, there aren't.
```

### 4. Hierarchical Clustering

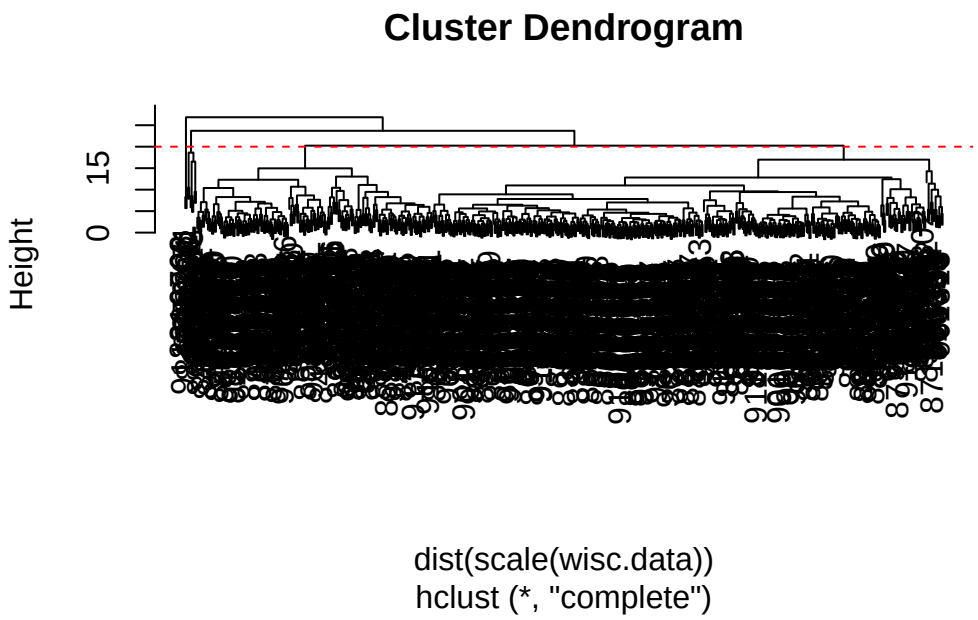
## Results of hierarchical clustering

Q10. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters?

```
> ans: 20
```

```r
# Scale the wisc.data data using the "scale()" function
wisc.hclust1 <- hclust(dist(scale(wisc.data)), method = "complete")
wisc.hclust <- hclust(dist(scale(wisc.data)))

plot(wisc.hclust)
abline(wisc.hclust, h=20, col="red", lty=2)
```




### Selecting number of clusters

You can also use the `cutree()` function and set k = 4 instead of h=height.

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

### Using different methods

> Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

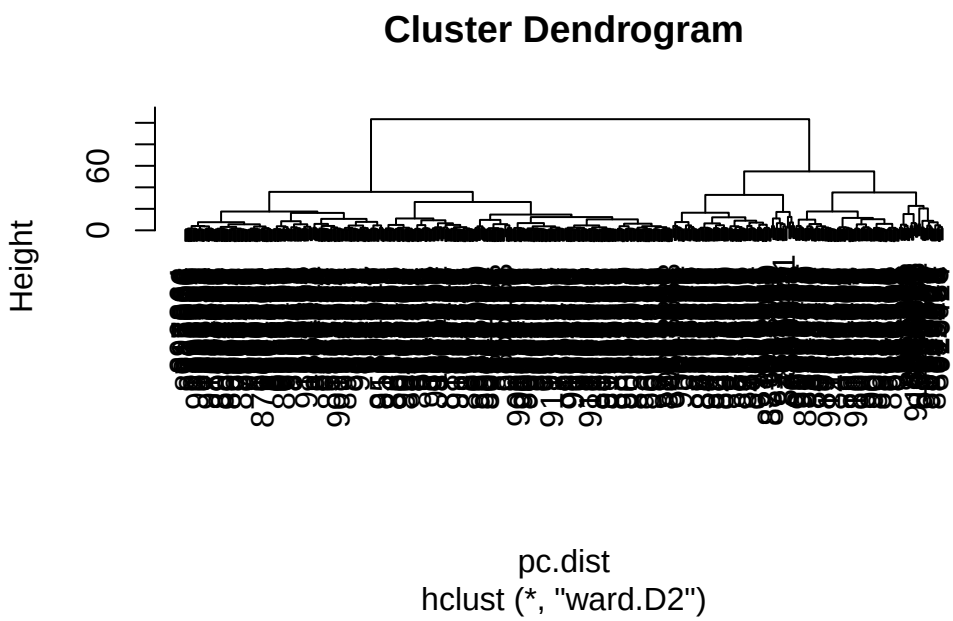
```
> ans: My favorite result is the "complete" version, because it's visually organized and co
```

### 5. Combining Methods

## Clustering on PCA results

Here we will take our PCA results and use those as an input for clustering. In other words our `wisc.pr$x` scores that we plotted above (the main output from PCA — how the data lie on our new pc axis/variables) and use a subset of those PC's that capture most variance as input for `hclust()`….

```
pc.dist<-dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(pc.dist, method = "ward.D2")
plot(wisc.pr.hclust)
```

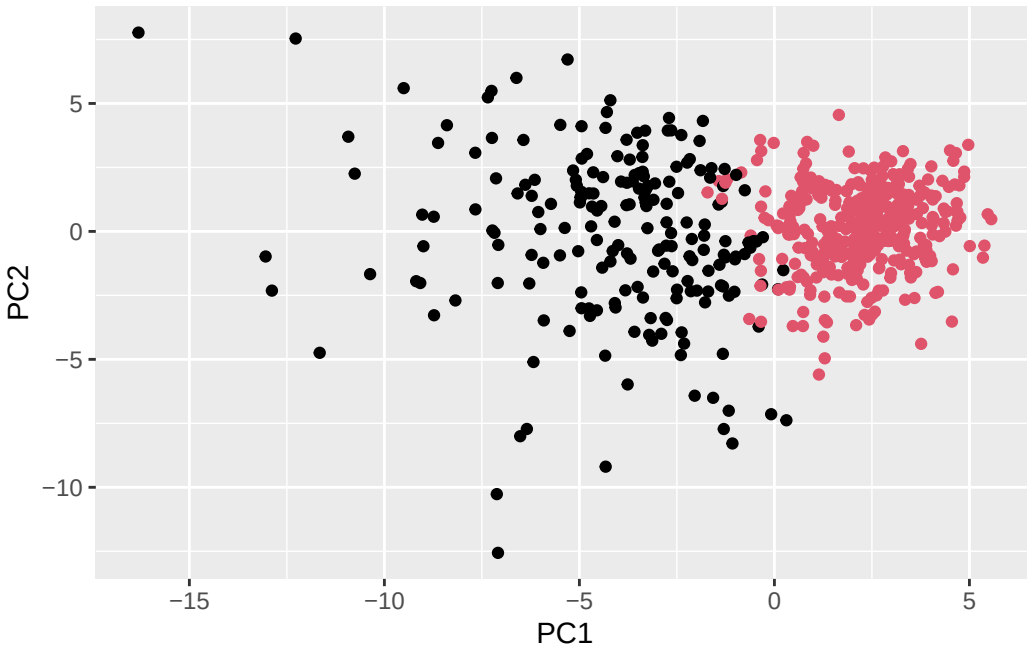



Cut the dendrogram/tree into groups/clusters

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

```r
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```




Q13. How does clustering into `grps` with values of 1 or 2 correspond with expert diagnosis?

```r
table(grps, diagnosis)
```

```
    diagnosis
grps   B   M
   1  24 179
   2 333  33
```

```r
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust.2 <- hclust(dist(wisc.pr$x[, 1:7]), method="ward.D2", members = NULL)

wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust.2, k=2)
```

```r
table(wisc.pr.hclust.clusters, diagnosis)
```

```
                       diagnosis
wisc.pr.hclust.clusters   B   M
                      1  28 188
                      2 329  24
```

```
# Compare to actual diagnoses

table(wisc.hclust.clusters, diagnosis)
```

```
                    diagnosis
wisc.hclust.clusters   B   M
                   1  12 165
                   2   2   5
                   3 343  40
                   4   0   2
```

> Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the table() function to compare the output of each model (wisc.hclust.clusters and wisc.pr.hclust.clusters) with the vector containing the actual diagnoses.

```
> ans: I think that while HC models are useful for basic visual assessments, PCA results na
```

### Sensistivity/Specificity

My clustering groups 1 are mostly "M" diagnosis (179) and my clustering 2 diagnosis are mostly "B" (333)…

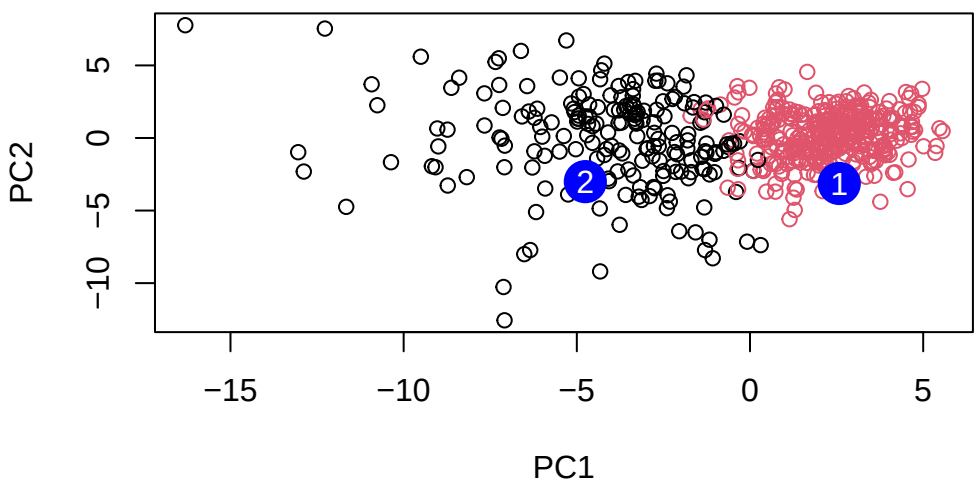
But….with the table, we have 24 false positives (eg: if you're in group 1, you'll falsely test positive for benign). True negatives 333. True positives are 179. 33 false negatives.

### Predictions

```
url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

```
           PC1       PC2        PC3        PC4       PC5        PC6        PC7
[1,]  2.576616 -3.135913  1.3990492 -0.7631950  2.781648 -0.8150185 -0.3959098
[2,] -4.754928 -3.009033 -0.1660946 -0.6052952 -1.140698 -1.2189945  0.8193031
            PC8       PC9       PC10      PC11      PC12      PC13     PC14
[1,] -0.2307350 0.1029569 -0.9272861 0.3411457  0.375921 0.1610764 1.187882
[2,] -0.3307423 0.5281896 -0.4855301 0.7173233 -1.185917 0.5893856 0.303029
          PC15       PC16        PC17        PC18        PC19       PC20
[1,] 0.3216974 -0.1743616 -0.07875393 -0.11207028 -0.08802955 -0.2495216
[2,] 0.1299153  0.1448061 -0.40509706  0.06565549  0.25591230 -0.4289500
           PC21       PC22       PC23       PC24        PC25         PC26
[1,]  0.1228233 0.09358453 0.08347651  0.1223396  0.02124121  0.078884581
[2,] -0.1224776 0.01732146 0.06316631 -0.2338618 -0.20755948 -0.009833238
             PC27        PC28         PC29         PC30
[1,]  0.220199544 -0.02946023 -0.015620933  0.005269029
[2,] -0.001134152  0.09638361  0.002795349 -0.019015820
```

```r
library(ggplot2)
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```




Q16. Which of these new patients should we prioritize for follow up based on your results? ans: Group 1.

## ## ChatGPT Usage

On Feb 1, 2026, at 10pm, I asked chat:

"wisc.pr.hclust.2 <- hclust(h, method="ward.D2", members = NULL) how to fix Error in if (is.na(n) || n > 65536L) stop("size cannot be NA nor exceed 65536") : missing value where TRUE/FALSE needed"

Chat responded with:

"That error isn't coming from hclust() itself — it's almost always because h (your distance object) contains NAs or is malformed." It then told me to check the class and if it wasn't a distance matrix, to use dist(wisc.pr).

==========